

Phase 5 Embedding / Estimator Results Summary (Updated)

Bold rows mark the current best result within each dataset/judge block. Primary criterion: lowest RMSE; secondary criterion: smaller absolute relative bias. The current headline point-estimation setup is BGE-M3 + learned concat-PCA + MIPS.

Mixed medical dialogue dataset (ACI 207 + MTS 142 + PriMock 48, n=397) | Judge: GPT-4o | V_b=0.6690, V_agent=0.8585, effect=+0.1895

Embedding	Est	V_hat	RelBias	RMSE	Dir%	95% CI	Cov	Note
SBERT	DM	0.7330	-14.62%	0.1357	100%	[0.697, 0.751]	no	
SBERT	MIPS	0.6636	-22.71%	0.2142	32%	[0.574, 0.719]	no	
SBERT	OffCEM	0.7256	-15.48%	0.1401	100%	[0.695, 0.747]	no	
OpenAI text-emb-3-small	DM	0.6884	-19.81%	0.1789	96%	[0.660, 0.700]	no	
OpenAI text-emb-3-small	MIPS	0.7494	-12.71%	0.1068	100%	[0.706, 0.793]	no	best GPT-4o
OpenAI text-emb-3-small	OffCEM	0.6921	-19.39%	0.1772	96%	[0.663, 0.703]	no	

Mixed medical dialogue dataset (n=397) | Judge: Claude Sonnet 4.6 | V_b=0.6178, V_agent=0.8413, effect=+0.2236

Embedding	Est	V_hat	RelBias	RMSE	Dir%	95% CI	Cov	Note
OpenAI text-emb-3-small	DM	0.6364	-24.35%	0.2111	91%	[0.609, 0.653]	no	
OpenAI text-emb-3-small	MIPS	0.6850	-18.58%	0.1526	99%	[0.635, 0.734]	no	
OpenAI text-emb-3-small	OffCEM	0.6427	-23.60%	0.2088	92%	[0.611, 0.656]	no	
MedCPT-Article	DM	0.6951	-17.38%	0.1577	100%	[0.665, 0.705]	no	
MedCPT-Article	MIPS	0.7525	-10.56%	0.0846	100%	[0.726, 0.789]	no	
MedCPT-Article	OffCEM	0.7000	-16.79%	0.1545	100%	[0.668, 0.709]	no	
BGE-M3	DM	0.7106	-15.53%	0.1352	100%	[0.673, 0.732]	no	
BGE-M3	MIPS	0.7659	-8.97%	0.0705	100%	[0.750, 0.793]	no	
BGE-M3	OffCEM	0.7123	-15.33%	0.1341	100%	[0.674, 0.734]	no	
MedCPT+BGE concat	DM	0.7162	-14.87%	0.1344	100%	[0.679, 0.733]	no	
MedCPT+BGE concat	MIPS	0.7619	-9.43%	0.0719	100%	[0.746, 0.795]	no	
MedCPT+BGE concat	OffCEM	0.7183	-14.62%	0.1329	100%	[0.680, 0.735]	no	
BGE-M3 + learned concat-PCA	DM	0.7025	-16.50%	0.1373	100%	[0.679, 0.730]	no	
BGE-M3 + learned concat-PCA	MIPS	0.7711	-8.34%	0.0674	100%	[0.751, 0.799]	no	best mixed/Sonnet
BGE-M3 + learned concat-PCA	OffCEM	0.7089	-15.74%	0.1344	100%	[0.683, 0.733]	no	

CounselBench | Judge: GPT-4 responder | V_b=0.5191, V_agent=0.6664, effect=+0.1473

Embedding	Est	V_hat	RelBias	RMSE	Dir%	95% CI	Cov	Note
BGE-M3	DM	0.5242	-21.34%	0.1382	71%	[0.488, 0.572]	no	
BGE-M3	MIPS	0.5479	-17.78%	0.1161	92%	[0.514, 0.588]	no	
BGE-M3	OffCEM	0.5243	-21.33%	0.1382	71%	[0.488, 0.572]	no	
BGE-M3 + learned concat-PCA	DM	0.5382	-19.24%	0.1297	88%	[0.504, 0.569]	no	
BGE-M3 + learned concat-PCA	MIPS	0.6008	-9.84%	0.0632	100%	[0.571, 0.639]	no	global best RMSE
BGE-M3 + learned concat-PCA	OffCEM	0.5384	-19.20%	0.1297	88%	[0.504, 0.569]	no	

CounselBench | Judge: LLaMA-3 responder | V_b=0.5191, V_agent=0.8580, effect=+0.3389

Embedding	Est	V_hat	RelBias	RMSE	Dir%	95% CI	Cov	Note
BGE-M3	DM	0.5978	-30.33%	0.2979	100%	[0.519, 0.610]	no	
BGE-M3	MIPS	0.6287	-26.72%	0.2307	100%	[0.571, 0.674]	no	
BGE-M3	OffCEM	0.5979	-30.31%	0.2979	100%	[0.519, 0.610]	no	
BGE-M3 + learned concat-PCA	DM	0.6011	-29.93%	0.2614	100%	[0.549, 0.637]	no	
BGE-M3 + learned concat-PCA	MIPS	0.6476	-24.52%	0.2091	100%	[0.607, 0.684]	no	best in subset
BGE-M3 + learned concat-PCA	OffCEM	0.6014	-29.90%	0.2613	100%	[0.549, 0.637]	no	

CounselBench | Judge: Gemini responder | V_b=0.5191, V_agent=0.6381, effect=+0.1191

Embedding	Est	V_hat	RelBias	RMSE	Dir%	95% CI	Cov	Note
BGE-M3	DM	0.5175	-18.91%	0.1197	53%	[0.488, 0.550]	no	
BGE-M3	MIPS	0.5084	-20.33%	0.1357	32%	[0.445, 0.558]	no	
BGE-M3	OffCEM	0.5176	-18.89%	0.1197	54%	[0.488, 0.550]	no	
BGE-M3 + learned concat-PCA	DM	0.5259	-17.60%	0.1167	54%	[0.497, 0.554]	no	best in subset
BGE-M3 + learned concat-PCA	MIPS	0.5086	-20.30%	0.1342	39%	[0.445, 0.566]	no	
BGE-M3 + learned concat-PCA	OffCEM	0.5258	-17.60%	0.1167	54%	[0.497, 0.554]	no	best in subset

Notes

- 1. Plain bootstrap CI coverage is still weak: every listed 95% bootstrap CI misses V_true(pi_agent). These tables therefore summarize point-estimation quality, not calibrated uncertainty.
- 2. CounselBench-GPT4 is the strongest transferred setting. CounselBench-LLaMA-3 and CounselBench-Gemini remain much harder despite learned embeddings.
- 3. DM and OffCEM are nearly identical in most cells, suggesting the residual correction is small under the current feature/density-ratio setup.